THE ZETA FUNCTION OF AN ELLIPTIC CURVE: SOLUTIONS

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Section 1 defines the zeta function of a curve over $\mathbb{F}_q$ as a generating function for its numbers of rational points, and shows that it is also an Euler product over the closed points of the curve. Section 2 computes the zeta function of an elliptic curve E from the action of Frobenius on the Tate module, proves the Riemann hypothesis for it, and deduces Hasse's bound $|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q}$. Section 3 shows that the zeta function, equivalently the single number $\#E(\mathbb{F}_q)$, is a complete isogeny invariant. Each problem is preceded by the definitions and results it uses.

Reading. [Sil09, §V.2], with [Sil09, §§III.7–III.8] for the Tate module, and [Poo06, §§2.2.1 and 3.4–3.5] for closed points and the Euler product. Three of the problems are exercises from [Sil09, Chapter V]: Problem 2.1 is Exercise 5.3, Problem 2.9 is Exercise 5.13 and Problem 3.4 is Exercise 5.4.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator.

Standing notation.

- A power $q = p^r$ of a prime p, and the Galois group $\mathrm{Gal}_{\mathbb{F}_q} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$.
- The automorphism $\sigma \in \mathrm{Gal}_{\mathbb{F}_q}$, $t \mapsto t^q$, of $\overline{\mathbb{F}}_q$, which generates $\mathrm{Gal}_{\mathbb{F}_q}$ topologically. Its fixed field in $\overline{\mathbb{F}}_q$ is $\mathbb{F}_q$, and that of σ^n is $\mathbb{F}_{q^n}$.
- In sections 2 and 3, an elliptic curve E over $\mathbb{F}_q$, given by a Weierstrass equation with coefficients in $\mathbb{F}_q$.
- The ring $\mathrm{End}(E)$ of the endomorphisms of E *defined over* $\mathbb{F}_q$.
- The q-power Frobenius endomorphism

$$\mathsf{F}\colon E \to E, \qquad (x, y) \mapsto (x^q, y^q).$$

- The integers

$$a \coloneqq q + 1 - \#E(\mathbb{F}_q), \qquad a_n \coloneqq q^n + 1 - \#E(\mathbb{F}_{q^n}) \quad (n \geqslant 1), \qquad a_0 \coloneqq 2,$$

  so that $a_1 = a$.
- For a prime $\ell \neq p$, the Tate module $T_\ell(E) = \varprojlim_n E[\ell^n](\overline{\mathbb{F}}_q)$, a free $\mathbb{Z}_\ell$-module of rank 2, and the map $\psi_\ell \in \mathrm{End}\big(T_\ell(E)\big)$ induced by $\psi \in \mathrm{End}(E)$.

How to use this. Allow about 25 minutes for §1, 35 for §2 and 20 for §3. In Problem 1.4 the heart is parts (2) and (3). Parts (5) and (6) can be done at home. Problems 2.1, 2.7 and 2.8 are the core of §2. Problem 2.9 is short, and Problem 2.10 is a computation to finish at home. In Problem 3.4, part (1) is quick, and the case analysis in parts (4) and (5) is the point of the problem. The `Magma` parts need no time in the session: do the hand work there and run the code afterwards, as a check on what you wrote. Every block returns in a few seconds.

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1. Zeta functions and closed points

Let C be a smooth projective curve over $\mathbb{F}_q$. The zeta function of C (Definition 1.2) packages the numbers of rational points over all the extensions of $\mathbb{F}_q$ into one power series. The exponential in it looks odd: the more obvious series is $\sum_n \#C(\mathbb{F}_{q^n})T^n$. Problem 1.4 explains the choice. It is [Poo06, Proposition 3.4.3 and Exercise 3.7]. In [Poo06] the ground field is perfect throughout, as $\mathbb{F}_q$ is.

Definition 1.1 (Closed points [Poo06, Definition 2.2.1])**.** A closed point P of C is an integral subvariety of dimension 0. Its residue field $\mathbb{F}_q(P)$ is a finite extension of $\mathbb{F}_q$, and its degree is $\deg P \coloneqq [\mathbb{F}_q(P) : \mathbb{F}_q]$.

Definition 1.2 (The zeta function)**.** The zeta function of C is

$$Z(C/\mathbb{F}_q; T) \coloneqq \exp\Big(\sum_{n\geqslant 1} \#C(\mathbb{F}_{q^n})\frac{T^n}{n}\Big) \in \mathbb{Q}[[T]],$$

so that $\#C(\mathbb{F}_{q^n})$ is the coefficient of T^{n-1} in $\frac{d}{dT}\log Z(C/\mathbb{F}_q; T)$.

Proposition 1.3 (Closed points are Galois orbits [Poo06, Remark 2.2.2])**.** *Sending a point of $C(\overline{\mathbb{F}}_q)$ to its image is a surjection from $C(\overline{\mathbb{F}}_q)$ onto the set of closed points of C, whose fibers are the $\mathrm{Gal}_{\mathbb{F}_q}$-orbits. The orbit over a closed point P has $\deg P$ elements: a point of $C(\overline{\mathbb{F}}_q)$ over P is an $\mathbb{F}_q$-embedding $\mathbb{F}_q(P) \hookrightarrow \overline{\mathbb{F}}_q$, and there are $\deg P$ of these.*

Problem 1.4 (The Euler product)**.**
For $d \geqslant 1$, let N_d be the number of closed points of C of degree d.

(1) Show that a point of $C(\overline{\mathbb{F}}_q)$ lies in $C(\mathbb{F}_{q^n})$ if and only if σ^n fixes it. Deduce that a $\mathrm{Gal}_{\mathbb{F}_q}$-orbit of size d is contained in $C(\mathbb{F}_{q^n})$ if $d \mid n$, and is disjoint from it otherwise.

Solution. Embed C in $\mathbb{P}^N$ over $\mathbb{F}_q$ and write a point of $C(\overline{\mathbb{F}}_q)$ as $Q = [x_0 : \cdots : x_N]$ with some $x_i = 1$. Then $\sigma^n Q = [x_0^{q^n} : \cdots : x_N^{q^n}]$, again with i-th coordinate 1, so $\sigma^n Q = Q$ if and only if every x_j lies in $\mathbb{F}_{q^n}$, the fixed field of σ^n, that is, $Q \in C(\mathbb{F}_{q^n})$. The coordinates of Q lie in some finite field $\mathbb{F}_{q^m}$, so the action of $\mathrm{Gal}_{\mathbb{F}_q}$ on the orbit of Q factors through $\mathrm{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$, which is generated by the image of σ. Hence the orbit is $\{Q, \sigma Q, \ldots, \sigma^{e-1}Q\}$, where σ^e generates the stabilizer of Q in $\langle\sigma\rangle$, and it has e elements. So $e = d$. Thus $\sigma^n Q = Q$ if and only if $d \mid n$, and since $\langle\sigma\rangle$ is abelian, every element of the orbit has the same stabilizer.

(2) Show that N_d is finite, and that

$$\#C(\mathbb{F}_{q^n}) = \sum_{d\mid n} dN_d \qquad \text{for every } n \geqslant 1.$$

Solution. By Proposition 1.3, a closed point of degree d is an orbit of size d, which by (1) lies in the finite set $C(\mathbb{F}_{q^d})$. So $N_d \leqslant \#C(\mathbb{F}_{q^d}) < \infty$. The set $C(\mathbb{F}_{q^n})$ is a union of orbits, namely of the orbits of size d for $d \mid n$, by (1). There are N_d of these, each with d elements, so $\#C(\mathbb{F}_{q^n}) = \sum_{d\mid n} dN_d$.

(3) Deduce the identity of formal power series

$$Z(C/\mathbb{F}_q; T) = \prod_P \big(1 - T^{\deg P}\big)^{-1} = \prod_{d\geqslant 1}(1 - T^d)^{-N_d},$$

the first product running over the closed points P of C. Why is the product well defined? Conclude that $Z(C/\mathbb{F}_q; T) \in 1 + T\mathbb{Z}[[T]]$, which is not obvious from the exponential.

SOLUTION. Since N_d is finite, only finitely many factors of the product are $\not\equiv 1 \pmod{T^m}$ for each m, namely those with $\deg P < m$. So the product converges in $\mathbb{Z}[[T]]$, and each factor $(1-T^d)^{-1} = 1 + T^d + T^{2d} + \cdots$ has integer coefficients. Taking logarithms and using $-\log(1-x) = \sum_{k\geqslant 1} x^k/k$,

$$\log \prod_{d\geqslant 1} (1-T^d)^{-N_d} = \sum_{d\geqslant 1} N_d \sum_{k\geqslant 1} \frac{T^{dk}}{k} = \sum_{n\geqslant 1} \Big(\sum_{d\mid n} dN_d \Big) \frac{T^n}{n} = \sum_{n\geqslant 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n},$$

with $n = dk$, so that $1/k = d/n$. Exponentiating gives the claim, and the product has coefficients in $\mathbb{Z}$ and constant term 1.

(4) Invert (2): show that $N_d = \frac{1}{d} \sum_{e\mid d} \mu(d/e)\, \#C(\mathbb{F}_{q^e})$, where μ is the Möbius function. So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other.

SOLUTION. Put $f(n) = \#C(\mathbb{F}_{q^n})$ and $g(d) = dN_d$. Then $f(n) = \sum_{d\mid n} g(d)$, and Möbius inversion gives $g(d) = \sum_{e\mid d} \mu(d/e) f(e)$.

(5) Take $C = \mathbb{P}^1$. Show that $Z(\mathbb{P}^1/\mathbb{F}_q; T) = 1/\big((1-T)(1-qT)\big)$. The closed points of $\mathbb{P}^1$ are the point at infinity, of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Deduce that the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$ is $\frac{1}{d} \sum_{e\mid d} \mu(d/e) q^e$. How many are there of degree 2 over $\mathbb{F}_2$, and which are they?

SOLUTION. **One: $t^2 + t + 1$.** $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1$, so

$$\log Z(\mathbb{P}^1/\mathbb{F}_q; T) = \sum_n (1+q^n) T^n/n = -\log(1-T) - \log(1-qT).$$

By (4) applied to $\mathbb{P}^1$, for $d > 1$ the point at infinity (the only closed point of $\mathbb{P}^1$ outside $\mathbb{A}^1$, of degree 1) does not count, and the extra $+1$ in $\#\mathbb{P}^1(\mathbb{F}_{q^e}) = q^e + 1$ contributes $\frac{1}{d} \sum_{e\mid d} \mu(d/e) = 0$. So for $d > 1$, and directly for $d = 1$, the number of monic irreducible polynomials of degree d is $\frac{1}{d} \sum_{e\mid d} \mu(d/e) q^e$. For $q = 2$ and $d = 2$: $\frac{1}{2}(4-2) = 1$, the polynomial $t^2 + t + 1$.

Remark. It is one closed point of degree 2, the orbit $\{\omega, \omega^2\}$ of a primitive cube root of unity in $\mathbb{F}_4$.

(6) Put $\zeta_C(s) \coloneqq Z(C/\mathbb{F}_q; q^{-s})$ and $NP \coloneqq \#\mathbb{F}_q(P)$. Show that $\zeta_C(s) = \prod_P \big(1 - NP^{-s}\big)^{-1}$, and compare with the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of the Riemann zeta function.

SOLUTION. Substitute $T = q^{-s}$: $T^{\deg P} = (q^{\deg P})^{-s} = NP^{-s}$, since $\mathbb{F}_q(P)$ has $q^{\deg P}$ elements. The closed points play the role of the primes, and NP that of $p = \#\mathbb{F}_p$.

Remark 1.5 (The Weil conjectures for curves)**.** Let C be a smooth projective curve of genus g over $\mathbb{F}_q$. The Weil conjectures (proved for curves by Weil) say:

(1) (Rationality) $Z(C/\mathbb{F}_q; T) = P(T)/\big((1-T)(1-qT)\big)$ with $P \in \mathbb{Z}[T]$ of degree $2g$ and $P(0) = 1$.
(2) (Functional equation) $Z\big(C/\mathbb{F}_q; 1/(qT)\big) = q^{1-g} T^{2-2g} Z(C/\mathbb{F}_q; T)$.
(3) (Riemann hypothesis) $P(T) = \prod_{i=1}^{2g} (1 - \alpha_i T)$ with $|\alpha_i| = \sqrt{q}$ for every i.

For $C = \mathbb{P}^1$, of genus 0, Problem 1.4(5) gives all three with $P = 1$. We prove them for $g = 1$ in section 2.

2. The zeta function of an elliptic curve

Problem 2.1 is [Sil09, Exercise 5.3].

Problem 2.1 (Traces and determinants)**.**
Let A be a square matrix over a field K of characteristic 0. We prove, in two ways, that

$$\exp\Big(\sum_{n\geqslant 1}\operatorname{tr}(A^n)\frac{T^n}{n}\Big)=\frac{1}{\det(1-AT)}\qquad\text{in } K[[T]]. \tag{2.1}$$

(1) Let $f,g\in 1+TK[[T]]$. Show that f and g are units of $K[[T]]$, so that their logarithmic derivatives f'/f and g'/g are defined, and that if $f'/f=g'/g$, then $f=g$. Where do you use that K has characteristic 0? Give a counterexample in characteristic p.

Solution. **Only to cancel n in $nh_n=0$.** A power series with constant term 1 is a unit: if $f=1-u$ with $u\in TK[[T]]$, then $f^{-1}=\sum_{k\geqslant 0}u^k$, which converges in $K[[T]]$ because $u^k\in T^kK[[T]]$. Put $h=f/g\in 1+TK[[T]]$. By the quotient rule,

$$h'=\frac{f'g-fg'}{g^2}=h\Big(\frac{f'}{f}-\frac{g'}{g}\Big)=0.$$

Writing $h=\sum_n h_nT^n$, this says $nh_n=0$ for every $n\geqslant 1$. Since K has characteristic 0, n is invertible in K, so $h_n=0$ for $n\geqslant 1$, and $h=h_0=1$. This is the only use of characteristic 0, and it is essential: in characteristic p, $f=1+T^p$ and $g=1$ have $f'=pT^{p-1}=0=g'$, so both logarithmic derivatives vanish, but $f\neq g$.

(2) Show that both sides of (2.1) lie in $1+TK[[T]]$, compute their logarithmic derivatives, and conclude with (1).

Solution. For $F\in TK[[T]]$, $\exp F$ has constant term 1, and the chain rule gives $(\exp F)'=F'\exp F$. So the left side lies in $1+TK[[T]]$, with logarithmic derivative $\sum_{n\geqslant 1}\operatorname{tr}(A^n)T^{n-1}$. The right side has constant term $1/\det(1)=1$. The matrix $M(T)=1-AT$ is invertible over $K[[T]]$, with inverse $\sum_{k\geqslant 0}A^kT^k$, and $M'(T)=-A$. So the logarithmic derivative of $\det(1-AT)^{-1}$ is

$$-\frac{\frac{d}{dT}\det(1-AT)}{\det(1-AT)}=-\operatorname{tr}\big((1-AT)^{-1}(-A)\big)=\operatorname{tr}\Big(\sum_{k\geqslant 0}A^{k+1}T^k\Big)=\sum_{n\geqslant 1}\operatorname{tr}(A^n)T^{n-1}.$$

The two agree, so the two sides are equal by (1).

(3) Give a second proof: triangularize A over an algebraic closure $\bar K$ of K, reduce (2.1) to the case of a 1×1 matrix $A=(\lambda)$, and prove that case with (1), applied over $\bar K$.

Solution. Over $\bar K$, $A=BUB^{-1}$ with U upper triangular, with diagonal entries the eigenvalues $\lambda_1,\dots,\lambda_m$ of A. Then $A^n=BU^nB^{-1}$, and U^n is upper triangular with diagonal entries λ_i^n, so $\operatorname{tr}(A^n)=\sum_i\lambda_i^n$. Likewise $1-AT=B(1-UT)B^{-1}$, so $\det(1-AT)=\prod_i(1-\lambda_iT)$. Since $\exp(F+G)=\exp F\exp G$, both sides of (2.1) are products over i of the corresponding sides for the 1×1 matrix (λ_i). For $A=(\lambda)$, both $\exp\big(\sum_n\lambda^nT^n/n\big)$ and $(1-\lambda T)^{-1}$ lie in $1+T\bar K[[T]]$, with logarithmic derivatives $\sum_{n\geqslant 1}\lambda^nT^{n-1}$ and $\lambda/(1-\lambda T)=\sum_{n\geqslant 1}\lambda^nT^{n-1}$. So they are equal by (1) over $\bar K$, which also has characteristic 0. Both sides of (2.1) have coefficients in K, so the identity in $\bar K[[T]]$ is the one we want.

Problem 2.7 is the proof of [Sil09, Theorems V.2.3.1 and V.2.4], arranged around Problem 2.1. In the following results, E is an elliptic curve over $\mathbb{F}_q$ and $\ell\neq p$ is a prime.

Proposition 2.2 (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *The Frobenius endomorphism* F *is purely inseparable of degree* q*.*

Corollary 2.3 ($1-\mathsf{F}$ is separable [Sil09, Corollary III.5.5])**.** *The endomorphism* $1-\mathsf{F}$ *is separable.*

Theorem 2.4 (Kernels of separable isogenies [Sil09, Theorem III.4.10(c)]). *A nonzero separable isogeny ψ satisfies $\#\ker\psi = \deg\psi$, where $\ker\psi$ is taken in $E(\overline{\mathbb{F}}_q)$.*

Proposition 2.5 (Determinant and trace on $T_\ell(E)$ [Sil09, Proposition III.8.6]). *For $\psi \in \mathrm{End}(E)$, $\det\psi_\ell = \deg\psi$ and $\operatorname{tr}\psi_\ell = 1 + \deg\psi - \deg(1-\psi)$.*

Theorem 2.6 (Endomorphisms act faithfully on $T_\ell(E)$ [Sil09, Theorem III.7.4]). *The map $\psi \mapsto \psi_\ell$ is an injective ring homomorphism $\mathrm{End}(E) \to \mathrm{End}\big(T_\ell(E)\big)$.*

Problem 2.7 (The zeta function of E).

(1) Show that $\sigma(Q) = \mathsf{F}(Q)$ for every $Q \in E(\overline{\mathbb{F}}_q)$. Deduce that the closed points of E of degree d are the F-orbits of size d in $E(\overline{\mathbb{F}}_q)$, and that $E(\mathbb{F}_{q^n}) = \ker(1-\mathsf{F}^n)$.

Solution. For $Q = (x_0, y_0)$ with $x_0, y_0 \in \overline{\mathbb{F}}_q$, $\mathsf{F}(Q) = (x_0^q, y_0^q) = \sigma(Q)$, and both fix O. So the $\mathrm{Gal}_{\mathbb{F}_q}$-orbits are the F-orbits (Problem 1.4(1): the orbit of Q is $\{Q, \sigma Q, \sigma^2 Q, \dots\}$), and Proposition 1.3 identifies them with the closed points. By Problem 1.4(1), $Q \in E(\mathbb{F}_{q^n})$ if and only if $\mathsf{F}^n(Q) = \sigma^n(Q) = Q$, that is, $Q \in \ker(1-\mathsf{F}^n)$.

(2) Show that F^n is the q^n-power Frobenius endomorphism of E over $\mathbb{F}_{q^n}$, and deduce that

$$\#E(\mathbb{F}_{q^n}) = \deg(1-\mathsf{F}^n) = \det(1-\mathsf{F}_\ell^n) = 1 - \operatorname{tr}(\mathsf{F}_\ell^n) + q^n.$$

Where does the argument use that $1-\mathsf{F}^n$ is separable?

Solution. **In $\#\ker(1-\mathsf{F}^n) = \deg(1-\mathsf{F}^n)$.** $\mathsf{F}^n(x,y) = (x^{q^n}, y^{q^n})$, and E is defined over $\mathbb{F}_{q^n}$. So Corollary 2.3, applied to E over $\mathbb{F}_{q^n}$, says that $1-\mathsf{F}^n$ is separable. It is nonzero, since its kernel $E(\mathbb{F}_{q^n})$ is finite while $E(\overline{\mathbb{F}}_q)$ is infinite. By Theorem 2.4 and (1), $\#E(\mathbb{F}_{q^n}) = \#\ker(1-\mathsf{F}^n) = \deg(1-\mathsf{F}^n)$. This is where separability is used, since for an inseparable isogeny $\#\ker$ is only the separable degree. By Proposition 2.5 and Theorem 2.6,

$$\deg(1-\mathsf{F}^n) = \det\big((1-\mathsf{F}^n)_\ell\big) = \det(1-\mathsf{F}_\ell^n).$$

For a 2×2 matrix M, $\det(1-M) = 1 - \operatorname{tr} M + \det M$, and $\det\mathsf{F}_\ell^n = (\det\mathsf{F}_\ell)^n = (\deg\mathsf{F})^n = q^n$ by Proposition 2.2.

(3) Taking $n=1$, show that the characteristic polynomial of F_ℓ is $T^2 - aT + q$. In particular it has integer coefficients and does not depend on ℓ.

Solution. For a 2×2 matrix the characteristic polynomial is $T^2 - (\operatorname{tr} M)T + \det M$. By (2) with $n=1$, $\operatorname{tr}\mathsf{F}_\ell = 1 + q - \#E(\mathbb{F}_q) = a$, and $\det\mathsf{F}_\ell = q$.

(4) Use Problem 2.1 to show that

$$Z(E/\mathbb{F}_q; T) = \frac{\det(1-\mathsf{F}_\ell T)}{(1-T)(1-qT)} = \frac{1-aT+qT^2}{(1-T)(1-qT)}.$$

By Problem 1.4, this is also $\prod_P (1-T^{\deg P})^{-1}$ over the closed points P of E.

Solution. By (2), $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \operatorname{tr}(\mathsf{F}_\ell^n)$, so, working in $\mathbb{Q}_\ell[[T]]$,

$$Z(E/\mathbb{F}_q;T) = \exp\Big(\sum_n \frac{T^n}{n}\Big)\exp\Big(\sum_n \frac{(qT)^n}{n}\Big)\exp\Big(-\sum_n \operatorname{tr}(\mathsf{F}_\ell^n)\frac{T^n}{n}\Big) = \frac{\det(1-\mathsf{F}_\ell T)}{(1-T)(1-qT)},$$

using Problem 2.1 over $K = \mathbb{Q}_\ell$ for the last factor. By (3),

$$\det(1-\mathsf{F}_\ell T) = T^2\det(T^{-1} - \mathsf{F}_\ell) = 1 - aT + qT^2.$$

(5) Verify the functional equation $Z\big(E/\mathbb{F}_q; 1/(qT)\big) = Z(E/\mathbb{F}_q;T)$ of Remark 1.5.

SOLUTION. Substituting $T \mapsto 1/(qT)$ and multiplying numerator and denominator by qT^2:

$$\frac{1 - a/(qT) + 1/(qT^2)}{\big(1 - 1/(qT)\big)\big(1 - 1/T\big)} = \frac{qT^2 - aT + 1}{(qT-1)(T-1)} = Z(E/\mathbb{F}_q; T),$$

which is Remark 1.5(2) with $g = 1$.

Problem 2.8 (The Riemann hypothesis and Hasse's bound)**.**
Let $\alpha, \beta \in \mathbb{C}$ be the roots of $T^2 - aT + q$, so that $1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T)$.

(1) For $r, s \in \mathbb{Z}$ with $s \neq 0$, show that $\det(r/s - \mathsf{F}_\ell) = \deg(r - s\mathsf{F})/s^2 \geqslant 0$. Deduce that $T^2 - aT + q$ is nonnegative on $\mathbb{R}$, and hence that $\beta = \bar{\alpha}$ and $|\alpha| = |\beta| = \sqrt{q}$. This is the Riemann hypothesis of Remark 1.5. With ζ_E as in Problem 1.4(6), show that every zero of $\zeta_E(s)$ satisfies $\mathrm{Re}(s) = 1/2$, and that $\zeta_E(s) = \zeta_E(1 - s)$.

SOLUTION. By Theorem 2.6, $(r - s\mathsf{F})_\ell = r - s\mathsf{F}_\ell$, so by Proposition 2.5

$$s^2 \det(r/s - \mathsf{F}_\ell) = \det(r - s\mathsf{F}_\ell) = \deg(r - s\mathsf{F}) \geqslant 0.$$

So the real polynomial $T^2 - aT + q = \det(T - \mathsf{F}_\ell)$ (Problem 2.7(3)) is $\geqslant 0$ on $\mathbb{Q}$, hence on $\mathbb{R}$ by continuity, and its discriminant $a^2 - 4q$ is $\leqslant 0$. Either the roots are non-real and complex conjugate, or there is a double real root. In both cases $\beta = \bar{\alpha}$, and $|\alpha|^2 = \alpha\beta = q$. The zeros of ζ_E are the s with $q^{-s} \in \{1/\alpha, 1/\beta\}$ (the denominator does not vanish there), so $q^{-\mathrm{Re}(s)} = |q^{-s}| = q^{-1/2}$. The substitution $s \mapsto 1 - s$ sends q^{-s} to $1/(q \cdot q^{-s})$, so $\zeta_E(1-s) = \zeta_E(s)$ is Problem 2.7(5).

Remark. Compare [Sil09, Remark V.2.5].

(2) Show that $a_n = \alpha^n + \beta^n$ for every $n \geqslant 0$. Deduce Hasse's bound

$$|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q},$$

and more generally $|a_n| \leqslant 2q^{n/2}$ for every $n \geqslant 1$.

SOLUTION. Taking logarithmic derivatives in Problem 2.7(4), written as

$$Z = \frac{(1 - \alpha T)(1 - \beta T)}{(1 - T)(1 - qT)},$$

gives $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \alpha^n - \beta^n$, that is, $a_n = \alpha^n + \beta^n$ for $n \geqslant 1$. And $a_0 = 2 = \alpha^0 + \beta^0$. Then $|a_n| \leqslant |\alpha|^n + |\beta|^n = 2q^{n/2}$ by (1), and $n = 1$ is Hasse's bound.

(3) Conversely, suppose only that there is a constant c with $|a_n| \leqslant c\, q^{n/2}$ for all $n \geqslant 1$. Show that the power series $\sum_{n \geqslant 1} a_n T^n = \frac{\alpha T}{1 - \alpha T} + \frac{\beta T}{1 - \beta T}$ converges for $|T| < q^{-1/2}$, and deduce that $|\alpha| = |\beta| = \sqrt{q}$. So the Riemann hypothesis is equivalent to a Hasse bound over every extension of $\mathbb{F}_q$, even with a worse constant.

SOLUTION. The series has coefficients bounded by $c\, q^{n/2}$, so it converges for $|T| < q^{-1/2}$ and defines a holomorphic function there. As formal power series, $\sum_n (\alpha^n + \beta^n) T^n = \alpha T/(1 - \alpha T) + \beta T/(1 - \beta T)$. If $\alpha \neq \beta$, the right side has poles at $1/\alpha$ and $1/\beta$, which do not cancel. If $\alpha = \beta$, it is $2\alpha T/(1 - \alpha T)$, with a pole at $1/\alpha$. A holomorphic function on the disc $|T| < q^{-1/2}$ has no poles there, so $|1/\alpha|, |1/\beta| \geqslant q^{-1/2}$, that is, $|\alpha|, |\beta| \leqslant \sqrt{q}$. Since $|\alpha||\beta| = q$, both are equalities.

Problem 2.9 is [Sil09, Exercise 5.13].

Problem 2.9 (A recurrence)**.**

(1) Show that the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geqslant 1$.

SOLUTION. $\#E(\mathbb{F}_q)$ determines a, hence $Z(E/\mathbb{F}_q;T)$ by Problem 2.7(4), hence every $\#E(\mathbb{F}_{q^n})$. Explicitly, it determines α and β, and $a_n = \alpha^n + \beta^n$ by Problem 2.8(2).

(2) Prove that

$$a_{n+2} = a_1 a_{n+1} - q a_n \qquad \text{for all } n \geqslant 0.$$

(Recall the convention $a_0 = 2$.)

SOLUTION. By Problem 2.8(2), $a_n = \alpha^n + \beta^n$ for $n \geqslant 0$. Since α is a root of $T^2 - aT + q$, $\alpha^{n+2} = a\alpha^{n+1} - q\alpha^n$, and likewise for β. Adding, $a_{n+2} = a\,a_{n+1} - q\,a_n$, with $a = a_1$.

Remark. In particular every a_n is an integer, computed from a_1 alone.

(3) Here is the recurrence on $y^2 = x^3 + x + 1$ over $\mathbb{F}_{101}$:

```
q := 101;  E := EllipticCurve([GF(q) | 1, 1]);          // y^2 = x^3 + x + 1
a := [TraceOfFrobenius(ChangeRing(E, GF(q^n))) : n in [1..6]];
a;                                  // [ -3, -193, 882, 16847, -139623, -1282678 ]
b := [2, a[1]];
for n in [1..5] do Append(~b, a[1]*b[n+1] - q*b[n]); end for;
b[2..7] eq a;                       // true
```

Here `TraceOfFrobenius(E)` is a. Compute a_2 by hand from $a_1 = -3$. Check Hasse's bound for a_6 over $\mathbb{F}_{101^6}$.

SOLUTION. **$a_2 = -193$. Hasse's bound holds for a_6.** $a_2 = a_1^2 - 2q = 9 - 202 = -193$, as Magma says, and the recurrence reproduces all six traces. For a_6:

$$2\sqrt{101^6} = 2 \cdot 101^3 = 2\,060\,602 \geqslant 1\,282\,678.$$

Problem 2.10 (A zeta function by hand)**.**
Let E be the elliptic curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_2$ (its discriminant is 1).

(1) List the rational points of $E(\mathbb{F}_2)$, and find a.

SOLUTION. **$E(\mathbb{F}_2) = \{O, (0,1), (1,0), (1,1)\}$. So $a = -1$.** For $x = 0$: $y^2 = 1$, so $y = 1$. For $x = 1$: $y^2 + y = 0$, so $y \in \{0, 1\}$. With O, $E(\mathbb{F}_2) = \{O, (0,1), (1,0), (1,1)\}$, so $\#E(\mathbb{F}_2) = 4$ and $a = 2 + 1 - 4 = -1$.

(2) Write down $Z(E/\mathbb{F}_2;T)$, and check the Riemann hypothesis for it directly.

SOLUTION. By Problem 2.7(4),

$$Z(E/\mathbb{F}_2;T) = \frac{1 + T + 2T^2}{(1-T)(1-2T)}.$$

The roots of $T^2 + T + 2$ are $\alpha, \beta = (-1 \pm \sqrt{-7})/2$, with $|\alpha|^2 = (1+7)/4 = 2 = q$.

(3) Use Problem 2.9 to predict $\#E(\mathbb{F}_4)$ and $\#E(\mathbb{F}_8)$. Check the first by listing the rational points of $E(\mathbb{F}_4)$, where $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ with $\omega^2 = \omega + 1$.

SOLUTION. **$\#E(\mathbb{F}_4) = 8$ and $\#E(\mathbb{F}_8) = 4$.** With $a_0 = 2$, $a_1 = -1$ and $q = 2$:

$$a_2 = (-1)(-1) - 2 \cdot 2 = -3, \qquad a_3 = (-1)(-3) - 2(-1) = 5.$$

So $\#E(\mathbb{F}_4) = 4+1+3 = 8$ and $\#E(\mathbb{F}_8) = 8+1-5 = 4$. To check the first: for $x = 0$, $y^2 = 1$ gives $y = 1$ only. For $x \neq 0$ put $y = xz$. Dividing by x^2, the equation becomes $z^2 + z = x + x^{-2}$. For $x = 1$ the right side is 0. For $x = \omega$ it is $\omega + \omega^{-2} = \omega + \omega = 0$ (as $\omega^3 = 1$). For $x = \omega^2$ it is $\omega^2 + \omega^{-4} = \omega^2 + \omega^2 = 0$. So in each case $z \in \{0, 1\}$, giving two rational points each, and $\#E(\mathbb{F}_4) = 1 + 1 + 3 \cdot 2 = 8$.

(4) Use Problem 1.4(4) to find the numbers N_1, N_2, N_3 of closed points of E of degree 1, 2 and 3. Deduce that every $\mathbb{F}_8$-rational point of E is $\mathbb{F}_2$-rational. Is $\#E(\mathbb{F}_8)$ close to the edge of the Hasse interval?

SOLUTION. **$N_1 = 4$, $N_2 = 2$, $N_3 = 0$. So $\#E(\mathbb{F}_8) = 4$ is the least Hasse allows.** $N_1 = \#E(\mathbb{F}_2) = 4$, $N_2 = \frac{1}{2}(8-4) = 2$ and $N_3 = \frac{1}{3}(4-4) = 0$. By Problem 1.4(2), $E(\mathbb{F}_8)$ is the union of the orbits of size 1 and 3, and there are none of size 3. So $E(\mathbb{F}_8) = E(\mathbb{F}_2)$. The Hasse interval over $\mathbb{F}_8$ is $|a_3| \leqslant 2\sqrt{8} \approx 5.66$, and $a_3 = 5$ is as large as it can be: $E(\mathbb{F}_8)$ is as small as an elliptic curve over $\mathbb{F}_8$ allows.

(5) Check the Euler product of Problem 1.4(3) against (2) up to the term in T^2.

SOLUTION. **Both are $1 + 4T + 12T^2$ modulo T^3.** Modulo T^3, the factors with $\deg P \geqslant 3$ are 1, so

$$\prod_P (1 - T^{\deg P})^{-1} \equiv (1-T)^{-4}(1-T^2)^{-2} \equiv (1 + 4T + 10T^2)(1 + 2T^2) \equiv 1 + 4T + 12T^2,$$

while $(1+T+2T^2)(1+3T+7T^2) \equiv 1+4T+12T^2$, using $1/\big((1-T)(1-2T)\big) = \sum_n (2^{n+1}-1)T^n$.

(6) Check, where `Places(E, d)` lists the closed points of degree d:

```
E := EllipticCurve([GF(2) | 1, 0, 0, 0, 1]);            // y^2 + xy = x^3 + 1
[#ChangeRing(E, GF(2^n)) : n in [1..8]];     // [ 4, 8, 4, 16, 44, 56, 116, 288 ]
[#Places(E, d) : d in [1..6]];               // [ 4, 2, 0, 2, 8, 8 ]
ZetaFunction(E);             // (2*$.1^2 + $.1 + 1)/(2*$.1^2 - 3*$.1 + 1)
```

SOLUTION. **They agree.** The counts agree with $2^n + 1 - a_n$ for the recurrence of Problem 2.9, and the numbers of closed points with (4) and, beyond it, with Problem 1.4(4): for instance $N_4 = \frac{1}{4}(16-8) = 2$. Magma's zeta function is ours, with both polynomials written in decreasing powers of its indeterminate, our T, and the denominator expanded: $2T^2 - 3T + 1 = (1-T)(1-2T)$.

3. Isogenous curves

Problem 3.4 is [Sil09, Exercise 5.4]. Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$, with q-power Frobenius endomorphisms F_1 and F_2 and traces $a(E_1)$, $a(E_2)$.

Definition 3.1 (Isogenies and Galois-equivariant maps)**.** We write $\mathrm{Hom}(E_1, E_2)$ for the isogenies $E_1 \to E_2$ *defined over* $\mathbb{F}_q$. The Galois group $\mathrm{Gal}_{\mathbb{F}_q}$ acts on each Tate module $T_\ell(E_i)$, and we write $\mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$ for the $\mathbb{Z}_\ell$-linear maps that commute with this action.

Theorem 3.2 (Tate's isogeny theorem [Sil09, Theorem III.7.7(a)])**.** *Let $\ell \neq p$ be prime. The natural map*

$$\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$$

is an isomorphism.

Corollary 3.3 (Isogenies form a free module [Sil09, Corollary III.7.5])**.** *The group* $\mathrm{Hom}(E_1, E_2)$ *is a free $\mathbb{Z}$-module of finite rank, being a subgroup of the free module of.*

Problem 3.4 (Isogenous curves have the same number of rational points)**.**

(1) Let $\varphi\colon E_1 \to E_2$ be a nonzero isogeny defined over $\mathbb{F}_q$. Show that $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$, and deduce that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

Solution. The isogeny φ is given by rational functions with coefficients in $\mathbb{F}_q$, and raising to the q-th power fixes those coefficients: $\varphi(x^q, y^q) = \varphi(x,y)^q$ coordinatewise, that is, $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$. Hence $\varphi \circ (1 - \mathsf{F}_1) = (1 - \mathsf{F}_2) \circ \varphi$, and taking degrees, which are multiplicative,

$$\deg \varphi \cdot \deg(1 - \mathsf{F}_1) = \deg(1 - \mathsf{F}_2) \cdot \deg \varphi.$$

Since $\deg \varphi \neq 0$, Problem 2.7(2) gives $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

(2) Deduce that $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$.

Solution. The zeta function depends only on q and $a = q + 1 - \#E(\mathbb{F}_q)$, by Problem 2.7(4).

For the converse, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$ (Problem 2.7(3)). Fix a prime $\ell \neq p$ and put $V_i \coloneqq T_\ell(E_i) \otimes \mathbb{Q}_\ell$.

(3) Show that a $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ commutes with the action of $\mathrm{Gal}_{\mathbb{F}_q}$ if and only if it intertwines $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$.

Solution. For $Q \in E_i(\overline{\mathbb{F}}_q)$ we have $\sigma(Q) = \mathsf{F}_i(Q)$, as in Problem 2.7(1). So σ acts on $T_\ell(E_i)$ by $\mathsf{F}_{i,\ell}$. A map commuting with $\mathrm{Gal}_{\mathbb{F}_q}$ therefore intertwines the Frobenius maps. Conversely, if λ commutes with σ, it commutes with every element of the subgroup $\langle\sigma\rangle$. The action of $\mathrm{Gal}_{\mathbb{F}_q}$ on each $E_i[\ell^n](\overline{\mathbb{F}}_q)$ factors through a finite quotient $\mathrm{Gal}(F/\mathbb{F}_q)$ for a finite extension F, which is generated by the image of σ. So λ commutes with the whole group modulo ℓ^n for every n, and hence on T_ℓ.

(4) Suppose $a^2 \neq 4q$. Show that $\mathsf{F}_{1,\ell}$ is not a scalar, that V_1 has a basis $v, \mathsf{F}_{1,\ell}v$, and that $V_1 \cong \mathbb{Q}_\ell[T]/(f)$ as $\mathbb{Q}_\ell[T]$-modules, with T acting by $\mathsf{F}_{1,\ell}$. Deduce that there is a nonzero $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwining the two Frobenius maps.

Solution. If $\mathsf{F}_{1,\ell} = c$ were a scalar, its characteristic polynomial would be $(T-c)^2$, with discriminant $a^2 - 4q = 0$. So some $v \in V_1$ is not an eigenvector, and then v and $\mathsf{F}_{1,\ell}v$ are linearly independent, hence a basis of the 2-dimensional space V_1. The map $\mathbb{Q}_\ell[T] \to V_1$, $g \mapsto g(\mathsf{F}_{1,\ell})v$, is therefore surjective. Its kernel contains f by Cayley–Hamilton, and $\mathbb{Q}_\ell[T]/(f)$ has dimension 2, so $V_1 \cong \mathbb{Q}_\ell[T]/(f)$. The same holds for V_2, so there is a $\mathbb{Q}_\ell[T]$-isomorphism $\lambda\colon V_1 \to V_2$, that is, a linear isomorphism with $\lambda \circ \mathsf{F}_{1,\ell} = \mathsf{F}_{2,\ell} \circ \lambda$. The image $\lambda(T_\ell(E_1))$ is a finitely generated $\mathbb{Z}_\ell$-submodule of V_2, so $\ell^k\lambda$ maps $T_\ell(E_1)$ into $T_\ell(E_2)$ for k large, and $\ell^k\lambda$ still intertwines and is nonzero.

(5) Suppose $a^2 = 4q$. Show that q is a square, that $a/2 \in \mathbb{Z}$, and that $\mathsf{F}_i = [a/2]$ in $\mathrm{End}(E_i)$ for $i = 1, 2$. Deduce that every $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwines the two Frobenius maps.

Solution. Since $a^2 = 4q$ is even, so is a. Thus $a/2 \in \mathbb{Z}$, and $q = (a/2)^2$ is the square of the integer $a/2 = \pm\sqrt{q}$. Now $f(T) = (T - a/2)^2$, and by Proposition 2.5

$$\deg\bigl(\mathsf{F}_i - [a/2]\bigr) = \det\bigl(\mathsf{F}_{i,\ell} - a/2\bigr) = f(a/2) = 0,$$

so $\mathsf{F}_i = [a/2]$. Then $\mathsf{F}_{i,\ell}$ is multiplication by $a/2$ on $T_\ell(E_i)$, and every $\mathbb{Z}_\ell$-linear map commutes with multiplication by a scalar.

(6) Use Theorem 3.2 to conclude that E_1 and E_2 are isogenous over $\mathbb{F}_q$.

SOLUTION. In either case, by (3), there is a nonzero element of $\mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$, so by Theorem 3.2 $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \neq 0$. As $\mathrm{Hom}(E_1, E_2) \cong \mathbb{Z}^\rho$ is free (Corollary 3.3), $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \cong \mathbb{Z}_\ell^\rho$, so $\rho \geqslant 1$ and there is a nonzero isogeny $E_1 \to E_2$ defined over $\mathbb{F}_q$.

Remark 3.5. Together with Problem 2.7(4), Problem 3.4 says that for elliptic curves E_1, E_2 over $\mathbb{F}_q$ the following are equivalent: E_1 and E_2 are isogenous over $\mathbb{F}_q$; $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$; $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$; and E_1 and E_2 have the same number of closed points of each degree (Problem 1.4(4)).

Looking ahead. By Problem 2.8(1), $\alpha = \sqrt{q}\, e^{i\theta}$ for a unique $\theta \in [0, \pi]$, and $a = 2\sqrt{q}\cos\theta$. Fix an elliptic curve over $\mathbb{Q}$, reduce it modulo every prime p of good reduction, and let θ_p be the angle of the reduction over $\mathbb{F}_p$. How are the θ_p distributed in $[0, \pi]$ as p varies? That is the Sato–Tate conjecture. The product over the primes of good reduction of the factors $(1 - a_p p^{-s} + p^{1-2s})^{-1}$, the reciprocals of the numerators of the ζ-functions of the reductions, is, up to finitely many factors, the L-function of the curve, the subject of the Birch and Swinnerton-Dyer conjecture.

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